

EduTracker ERP: Developer Onboarding & Installation Guide

This guide is for engineers setting up a fresh development environment after pulling the repository.

Step 1: System Requirements

Ensure you have the following installed: 1. **Node.js** (v18 or higher) 2. **PostgreSQL** (v16 or higher) 3. **Git** 4. **VS Code** (Recommended)

Step 2: Clone & Install Dependencies

1. **Pull the code:** `bash git clone <your-repo-url> cd eduTracker`
 2. **Install Frontend Dependencies:** `bash npm install`
 3. **Install Backend Dependencies:** `bash cd backend npm install cd ..`
-

Step 3: Environment Configuration

1. **Root .env:** Create a .env file in the root directory: `env NEXT_PUBLIC_API_URL="http://localhost"`
 2. **Backend .env:** Create a .env file in the backend directory: `env DATABASE_URL="postgresql://postgres:YOUR_PASSWORD@localhost:5432/edutracker?schema=public" ACCESS_TOKEN_SECRET="generate-a-random-long-string" REFRESH_TOKEN_SECRET="generate-a-random-long-string" PORT=5000 NODE_ENV=development`
-

Step 4: Database Setup

1. **Generate Prisma Client:** `bash cd backend npx prisma generate`
 2. **Run Migrations:** Apply the database schema to your local PostgreSQL: `bash npx prisma migrate dev`
 3. **Seed Initial Data:** Populate the database with a default Admin user and initial settings: `bash npx prisma db seed`
-

Step 5: Start Development Servers

1. **Start Backend:** `bash` # Inside the 'backend' folder `npm run dev`
 2. **Start Frontend:** `bash` # Inside the root 'eduTracker' folder `npm run dev`
-

Step 6: Verify Installation

1. Open `http://localhost:3000` in your browser.
 2. Login with the default seed credentials:
 - **Email:** `admin@edutracker.com`
 - **Password:** `admin123`
 3. Check if you can see the **Dashboard**.
-

Key Maintenance Commands

- **Update Schema:** After changing `schema.prisma`, run `npx prisma migrate dev --name <description>`.
 - **Database Studio:** Run `npx prisma studio` to view data in a browser.
-

EduTracker ERP - Engineering Excellence.